Nimmala Sree Prabhath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.25
Tirumala Junior College	Class XII (Board of Intermediate Education Andhra Pradesh)	2021	97.60%
Sri Chaitanya Techno (EM) School	Class X (Board of Secondary Education Andhra Pradesh)	2019	9.80%

SCHOLASTIC ACHIEVEMENTS

Achieved a perfect 10/10 GPA in the first semester of M.Tech

Secured an **A+** grade in **Analog IC Design**

SKILLS AND EXPERTISE

Tools and Software : Cadence Virtuoso | LTspice | XILINX Vivado | Icarus Verilog

Programming and HDL : Verilog | MATLAB | C

Soft Skills : Analytical Problem-Solving | Technical Communication | Team Collaboration

PROJECTS

Designed Two-Stage Operational Transconductance Amplifier (OTA) | Cadence Virtuoso , 45nm CMOS Technology

- Designed a **two-stage Miller-compensated OTA** using I/O (thick-oxide) devices for extended voltage headroom , meeting the following performance **specifications:**
- Performed DC, AC, and transient analysis to evaluate gain, bandwidth, phase margin, slew rate, and power consumption meeting the following performance **specifications:**
- Achieving **DC Gain 50 dB** (320 v/v),Gain-Bandwidth Product (GBW) of 30 MHz, **Phase Margin of 52**, Slew Rate 20 uv/s, Input Common-Mode Range 0.8v to 1.6v, Load Capacitance 2pF,and Power Consumption 300uW

Designed Bandgap Reference (BGR) Circuit Design | Cadence Virtuoso, 45nm CMOS Technology

- Designed and simulated two BGR topologies current-mirror based and Op-amp based in Cadence using **PTAT/CTAT voltage summation**, achieving a temperature-compensated output voltage of **1.25v**, with TC stability verified across -40 to 125 degrees (centigrade).
- Incorporated startup circuitry in both designs to ensure reliable power-up, and characterized supply sensitivity (PSRR) via DC sweep.

Designed Low Dropout Regulator (LDO) | Cadence Virtuoso, 45nm CMOS Technology

- Designing a basic LDO regulator with **1.8v input, 1.5v output, 300mv dropout**, targeting a 10mA load current, reusing the Bandgap reference 1.25v topology from the BGR project for voltage reference generation.
- Implementing an error amplifier and PMOS pass transistor with resistive feedback network,targeting **45** and **60 phase margin** for loop stability,and verifying performance through DC operating point, AC loop-gain, and the transient (load/line regulation) simulations.

AWARDS AND ACHIEVEMENTS

- **Academic Excellence** Awardee for securing the **Highest GPA** across all M.Tech branches in the **Electrical Department .** Presented by **Prof. B. S. Murty, Director**, IIT Hyderabad.
- Represented school at the **District-level chess championship** and Won a **Medal** in **college-level chess**

RELEVANT COURSES

Analog Domain : Analog IC Design , Analog IC Design Lab .

Digital Domain : CMOS Digital Design , Digital IC Design , Digital IC Design Lab .

Devices and Fabrication: Semiconductor device modelling , VLSI Technology.

CAMPUS ACTIVITIES AND LEADERSHIP

Teaching Assistantship:

- TA for Digital Systems Design course Under **Prof. Gajendranath Chaudhury .**
- TA for Electronic Devices and Circuits Lab Under **Prof. Gajendranath Chaudhury.**
- TA for Basic Electric Circuits Course **NPTEL , IITM**

POSITIONS OF RESPONSIBILITY

***Teaching Assistant* | *Digital Systems Design* (EE1201)**

- Mentored B.Tech first-year students through intuitive explanations, visual illustrations, and engaging problem-solving by conducting tutoring sessions.

***Teaching Assistant* | *Electronic Devices and Circuit laboratory* (EE2301)**

- Guided B.Tech second-year students through hands-on experiments and circuit troubleshooting, bridging theoretical concepts with practical circuit implementation.

EXTRACURRICULAR ACTIVITIES

Cricket | Chess | Story Writing